Solutions

1. (16 points) Suppose a system of linear equations $A\mathbf{x} = \mathbf{b}$ with variables x_1, x_2, x_3, x_4, x_5 is reduced to

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 2 & 0 & 4b_1 - b_3 \\ 0 & 1 & 1 & 1 & 0 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 1 & b_3 - b_1 - b_2 \\ 0 & 0 & 0 & 0 & 0 & b_4 - 3b_1 \\ 0 & 0 & 0 & 0 & 0 & b_5 - 2b_1 + b_2 \end{array}\right], \text{ where } \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}.$$

Answer the following questions about this system.

(1.1) Identify the leading variable(s) in the system. **Select ALL that apply.**

☒ (a) x_1 ☒ (b) x_2 ☐ (c) x_3

☐ (d) x_4 ☒ (e) x_5 ☐ (f) There are no leading variables

(1.2) Identify the free variable(s) in the system. **Select ALL that apply.**

☐ (a) x_1 ☐ (b) x_2 ☒ (c) x_3

☒ (d) x_4 ☐ (e) x_5 ☐ (f) There are no free variables

(1.3) Is it possible that $A\mathbf{x} = \mathbf{b}$ is consistent? ● yes ○ no

If yes, list the condition(s) on b_1, b_2, b_3, b_4, b_5: $b_4 - 3b_1 = 0$

$b_5 - 2b_1 + b_2 = 0$

If no, justify:

(1.4) For which of the following $\mathbf{b}$, if any, is the system $A\mathbf{x} = \mathbf{b}$ consistent? **Select ALL that apply.**

☐ (a) $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \\ 2 \end{bmatrix}$ $3 - 3(1) = 0$ ✓ $2 - 2(1) + 1 \neq 0$

☒ (b) $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}$ $3 - 3(1) = 0$ ✓ $0 - 2(1) + 2 = 0$ ✓

☐ (c) $\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 4 \\ 1 \\ 1 \end{bmatrix}$ $1 - 3(1) \neq 0$

☐ (d) $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ 1 \\ 1 \end{bmatrix}$ $1 - 3(3) \neq 0$

☒ (e) $\mathbf{b} = \begin{bmatrix} 0 \\ 2 \\ -5 \\ 0 \\ -2 \end{bmatrix}$ $0 - 3(0) = 0$ ✓ $-2 - 2(0) + 2 = 0$

☒ (f) $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

☐ (g) $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 3 \\ 1 \end{bmatrix}$ $3 - 3(1) = 0$ ✓ $1 - 2(1) + 2 \neq 0$

☐ (h) none

(1.5) Is it possible that $A\mathbf{x} = \mathbf{b}$ has a unique solution? ○ yes ● no

If yes, list the condition(s) on b_1, b_2, b_3, b_4, b_5?

If no, justify:

There are free variables. The system is either inconsistent or it will be consistent with infinitely many solutions.

2. (12 points) In each of the following cases, decide if the linear system corresponding to the augmented matrix is consistent or inconsistent by filling in the bubble corresponding to your answer. If the system is consistent, write the **vector form of the solution**.

(2.1) $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$ ● consistent ○ inconsistent

If consistent, solution $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \\ -1 \\ 0 \end{bmatrix}$

(2.2) $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$ ○ consistent ● inconsistent

If consistent, solution $\mathbf{x} =$

(2.3) $\left[\begin{array}{ccccc|c} 1 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & -3 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$ ● consistent ○ inconsistent

$$\begin{aligned} x_1 + x_2 - 2x_4 &= 1 \\ x_3 + x_4 &= -3 \\ x_5 &= 0 \end{aligned} \Rightarrow \begin{aligned} x_1 &= 1 - x_2 + 2x_4 \\ x_2 &= x_2 \\ x_3 &= -3 - x_4 \\ x_4 &= x_4 \\ x_5 &= 0 \end{aligned}$$

If consistent, solution $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ -3 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}$

(2.4) $\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ ◐ consistent ○ inconsistent

$$\begin{aligned} x_1 + x_2 &= 0 \\ x_3 &= 0 \end{aligned} \Rightarrow \begin{aligned} x_1 &= -x_2 \\ x_2 &= x_2 \\ x_3 &= 0 \end{aligned}$$

If consistent, solution $\mathbf{x} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

3. (18 points) Let $A = \begin{bmatrix}1 & 2\\ -2 & 3\end{bmatrix}$, $B = \begin{bmatrix}4 & 2\\ 1 & 0\end{bmatrix}$, $C = \begin{bmatrix}-1 & 2\\ 1 & 0\\ -2 & 1\end{bmatrix}$, $\mathbf{u} = \begin{bmatrix}1\\ 3\end{bmatrix}$, $\mathbf{w} = \begin{bmatrix}-2\\ 3\\ 2\end{bmatrix}$. Compute the following expressions when they are defined or state they are undefined and explain why.

(3.1) $3A + B^T$ $= 3\begin{bmatrix}1 & 2\\ -2 & 3\end{bmatrix} + \begin{bmatrix}4 & 1\\ 2 & 0\end{bmatrix} = \begin{bmatrix}3 & 6\\ -6 & 9\end{bmatrix} + \begin{bmatrix}4 & 1\\ 2 & 0\end{bmatrix} = \begin{bmatrix}7 & 7\\ -4 & 9\end{bmatrix}$

(3.2) $\overset{(2\times2)}{A} + BC^T$

$(2\times2)(2\times3)$

(2×3)

Undefined, A is (2×2) and BC^T is (2×3)
cannot add matrices that are different sizes.

(3.3) CA $= \begin{bmatrix}-1 & 2\\ 1 & 0\\ -2 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ -2 & 3\end{bmatrix} = \begin{bmatrix}-5 & 4\\ 1 & 2\\ -4 & -1\end{bmatrix}$

$(3\times2)(2\times2)$

$=$

(3.4) AC

$(2\times2)(3\times2)$

$\neq$

Undefined, the # of columns of A is not equal to the # of rows of C. (Inner dimensions do not match)

(3.5) $C\mathbf{u}$ $= \begin{bmatrix}-1 & 2\\ 1 & 0\\ -2 & 1\end{bmatrix}\begin{bmatrix}1\\ 3\end{bmatrix} = 1\begin{bmatrix}-1\\ 1\\ -2\end{bmatrix} + 3\begin{bmatrix}2\\ 0\\ 1\end{bmatrix} = \begin{bmatrix}5\\ 1\\ 1\end{bmatrix}$

$(3\times2)(2\times1)$

$=$

(3.6) $C\mathbf{w}$

$(3\times2)(3\times1)$

$\neq$

Undefined, the # of columns of C is not equal to the # of rows of $\vec{w}$. (Inner dimensions do not match)

4. (10 points) Identify whether each one of the given sets is linearly dependent or linearly independent by filling in the bubble corresponding to your answers Explanations are not necessary. *Hint:* all can be done by inspection.

(4.1) $\left\{ \begin{bmatrix} -2 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

(4.2) $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}$ ○ linearly dependent ○ linearly independent

(4.3) $\left\{ \begin{bmatrix} -4 \\ 2 \end{bmatrix}, \begin{bmatrix} 8 \\ -4 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

(4.4) $\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

(4.5) $\left\{ \begin{bmatrix} 4 \\ 4 \end{bmatrix} \right\}$ ○ linearly dependent ● linearly independent

(4.6) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

(4.7) $\left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix} \right\}$ ○ linearly dependent ● linearly independent

(4.8) $\left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

(4.9) $\left\{ \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$ ○ linearly dependent ○ linearly independent

(4.10) $\left\{ \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ ● linearly dependent ○ linearly independent

5. (8 points) Let $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & -2 \\ 3 & 4 & a \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & -2 \\ 3 & 4 & a+1 \end{bmatrix}$.

(5.1) Find the value(s) of a so that the vectors $\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$, and $\begin{bmatrix} 2 \\ -2 \\ a \end{bmatrix}$ are linearly dependent.

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -2 & 0 \\ 3 & 4 & a & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & -2 & a-6 & 0 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & a-10 & 0 \end{array}\right]$$

$\Rightarrow a - 10 = 0$

$\Rightarrow a = 10$

(5.2) For the value of a you found in 5.1, is the matrix A nonsingular (i.e., invertible)? ○ yes ● no

(5.3) For the value of a you found in 5.1, is the matrix B nonsingular? ● yes ○ no

6. (5 points) For each true/false question, completely fill in the answer bubble that corresponds to your answer. Explanations are not necessary.

(6.1) ○ T ● F It is possible for a (5×5) system of linear equations to have exactly 5 solutions.

(6.2) ○ T ● F If A is a (4×5) matrix and $\mathbf{0}$ is the zero vector in $\mathbb{R}^5$, then it is possible that $A\mathbf{x} = \mathbf{0}$ has a unique solution.

(6.3) ● T ○ F If A is $(n \times n)$ and nonsingular then $A\mathbf{x} = \mathbf{b}$ has a unique solution in $\mathbb{R}^n$.

(6.4) ○ T ● F If A is a (5×4) matrix and $\mathbf{b}$ is any vector in $\mathbb{R}^4$, then $A\mathbf{x} = \mathbf{b}$ always has at least one solution.

(6.5) ● T ○ F If A is an $(m \times n)$ matrix and C is an $(n \times p)$ matrix then $(AC)^T = C^T A^T$.

7. (10 points) Let $A = \begin{bmatrix} 1 & 0 & 1 \\ -4 & 1 & -4 \\ 2 & 4 & 3 \end{bmatrix}$.

(7.1) The matrix A **is invertible**. Find its inverse A^{-1}. Show all your work and write your final answer in the space provided.

$$A^{-1} = \begin{bmatrix} 19 & 4 & -1 \\ 4 & 1 & 0 \\ -18 & -4 & 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ -4 & 1 & -4 & 0 & 1 & 0 \\ 2 & 4 & 3 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2 \to R_2 + 4R_1 \\ R_3 \to R_3 - 2R_1 \end{array} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 4 & 1 & 0 \\ 0 & 4 & 1 & -2 & 0 & 1 \end{array}\right]$$

$$R_3 \to R_3 - 4R_2 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & -18 & -4 & 1 \end{array}\right]$$

$$R_1 \to R_1 - R_3 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 19 & 4 & -1 \\ 0 & 1 & 0 & 4 & 1 & 0 \\ 0 & 0 & 1 & -18 & -4 & 1 \end{array}\right]$$

(7.2) **Use your answer from 7.1** to solve the system $A\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} 19 & 4 & -1 \\ 4 & 1 & 0 \\ -18 & -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 3 \\ -12 \end{bmatrix}$$

8. (5 points) Let A and B be the matrices such that $A^{-1} = \begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 3 \\ 0 & -2 & -5 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \\ -1 & 1 & 1 \end{bmatrix}$.
What is $(AB)^{-1}$? **Select <u>one</u> answer.** No work required.

(A) $(AB)^{-1}$ is not defined.

(B) $\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & -4 & -9 \\ 0 & -3 & -7 \\ -1 & -2 & -6 \end{bmatrix}$

(D) Not enough information to calculate $(AB)^{-1}$.

(E) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -2 \\ 4 & 3 & -5 \end{bmatrix}$

(F) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(G) $\begin{bmatrix} 1 & -1 & 8 \\ 0 & 1 & 6 \\ 0 & -2 & -5 \end{bmatrix}$

(H) $\begin{bmatrix} -3 & 4 & 8 \\ -3 & 4 & 5 \\ 5 & -7 & -9 \end{bmatrix}$

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 3 \\ 0 & -2 & -5 \end{bmatrix}$$

9. (4 points) Which of the following matrices are symmetric? Select **ALL answers that apply**. No credit will be given if all answers are left blank or all answers are selected. No work required.

☒ (a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

☐ (b) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

☐ (c) $\begin{bmatrix} 1 & -2 & -3 \\ 2 & 3 & -4 \\ 3 & 4 & 1 \end{bmatrix}$

☒ (d) $\begin{bmatrix} 1 & 3 & 2 \\ 3 & 6 & 8 \\ 2 & 8 & 7 \end{bmatrix}$

☐ (e) $\begin{bmatrix} -3 & 7 & 4 \\ 7 & -3 & 5 \\ 4 & 6 & -3 \end{bmatrix}$

☒ (f) $\begin{bmatrix} 2 & -1 & 3 \\ -1 & 7 & -4 \\ 3 & -4 & 0 \end{bmatrix}$

☐ (g) $\begin{bmatrix} 1 & 3 & 1 \\ 3 & 1 & 3 \end{bmatrix}$

☐ (h) $\begin{bmatrix} 2 & -3 & 4 \\ -1 & 7 & -3 \\ 4 & -1 & 2 \end{bmatrix}$

10. (12 points) Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Indicate if the subset W of $\mathbb{R}^3$ is a subspace of $\mathbb{R}^3$ by filling in the bubble corresponding to your answers. In the cases where W is **not** a subspace, give a **specific** counterexample that **clearly shows which property fails**. In the cases where W **is** a subspace of $\mathbb{R}^3$, give a geometric description of W. (You do NOT need to verify in general.)

(10.1) $W = \{\mathbf{x} : 3x_1 + x_2 - 4x_3 = 0\}$ ○ Not a subspace of $\mathbb{R}^3$ ● W is a subspace of $\mathbb{R}^3$

Counterexample OR geometric description of W:

$\underbrace{\text{Plane through origin}}_{\text{must be stated}}$ with normal vector $\vec{n} = \langle 3, 1, -4 \rangle$

(10.2) $W = \{\mathbf{x} : \ x_1 \geq 0\}$ ○ Not a subspace of $\mathbb{R}^3$ ○ W is a subspace of $\mathbb{R}^3$

Counterexample OR geometric description of W:

$\vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \in W$ but if $a = -1$, then $a\vec{x} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} \notin W$ since $x_1 < 0$

(10.3) $W = \{\mathbf{x} : x_3 = x_2 = 2x_1\}$ ○ Not a subspace of $\mathbb{R}^3$ ● W is a subspace of $\mathbb{R}^3$

Counterexample OR geometric description of W:

$\vec{x} = \begin{bmatrix} x_1 \\ 2x_1 \\ 2x_1 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

$\underbrace{\text{line through the origin}}_{\text{must be stated}}$ with direction vector $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

(10.4) $W = \{\mathbf{x} : \ x_2 x_3 = 0\}$ ● Not a subspace of $\mathbb{R}^3$ ○ W is a subspace of $\mathbb{R}^3$

Counterexample OR geometric description of W:

Let $\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \in W$ and $\vec{y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \in W$

since $0(1) = 0$ Since $1(0) = 0$

$\vec{x} + \vec{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \notin W$ since $1(1) = 1 \neq 0$